

RAHUL DEBNATH

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Full-Stack Developer with experience building scalable web applications and RESTful APIs. Strong foundation in software engineering, databases, and problem-solving. Co-author of an IEEE-published research paper on quantum-inspired optimization algorithms.

EDUCATION

- B.Tech in Computer Science & Engineering (MAKAUT)** | B.P. Poddar Institute of Management & Technology, Kolkata
CGPA: 7.8 (till 7th Sem) **Expected: 2026**

SKILLS

- Languages:** C, Java, Python, JavaScript
- Web Technologies:** HTML, CSS, Node.js, React.js, Tailwind CSS
- Databases:** SQL, Mongo DB
- Developer Tools:** VS Code, Git, GitHub
- Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems

EXPERIENCE

Intern | CODTECH IT SOLUTIONS PRIVATE LIMITED | Remote

February–April 2025

- Developed and deployed frontend and full-stack web applications using HTML, CSS, JavaScript, React.js, Node.js, Mongo DB, Redis
- Built a real-time Chat Application and Integrated Google Gemini API for enhanced project functionalities.

PROJECTS

- AI Chat Application** | [view](#) *January–April 2025*
 - Built an AI-powered chat application using React.js and Node.js, focused on smart and engaging conversation through AI integration.
 - Tech Used:** React.js, Node.js, MongoDB, Redis, Socket.io, Tailwind-CSS, Google Gemini API
- Movie Ticket Booking Application** | [view](#) *May–July 2025*
 - A dynamic movie booking app with real-time details, responsive UI, seamless ticketing and robust error handling.
 - Tech Used:** React.js, Node.js, Mongo DB, Tailwind-CSS, Clerk, OMDB API
- Personal AI Assistant** | [view](#) *March–April 2025*
 - Upgraded a voice assistant with React.js and Google Gemini 2.0 API for dynamic AI responses.
 - Added voice input/output using Web Speech API; now handles open-ended prompts, weather and more.
 - Tech Used:** React.js, JavaScript, Google Generative AI (Gemini), Web Speech API

PUBLICATION

- A. Kumari, P. Bhowmik, S. Poddar, R. Debnath, A. Dutta, R. Majumder, and R. K. Pal, “QIACO-VQSF: A Hybrid Classical–Quantum ACO Framework for the Travelling Salesman Problem,”** accepted for presentation at the IEEE Computer Society Annual Symposium on VLSI (ISVLSI 2026), 2026. | [view](#)
- Explored quantum-inspired search mechanisms, probabilistic state representations, and hybrid optimization strategies combining classical metaheuristics with quantum-inspired techniques.
- Explored quantum computing for VLSI and engineering optimization.

CERTIFICATION

Meta Backend Developer Professional | Completed Meta Backend Developer Professional Certification, gaining hands-on expertise in Python, APIs, databases and backend system design. | [view](#)